Alessandro Perini

PhD student in Industrial Engineering, University of Rome Tor Vergata

Education

2024-Present	University of Rome Tor Vergata, PhD in Industrial Engineering Innovative Doctoral Fund DM630, funded by the Ministry for Universities and Research, NextGenerationEU (EU), and Sensor Medica (Italy). "*Development of wearable robotic systems*." Supervisor: Dr. Matteo Russo
2022-2024	University of Rome Tor Vergata, MSc in Mechatronics Engineering, cum laude Thesis: "*Development of a tendon-driven hand exoskeleton.*"
2019-2022	University of Roma Tre, BSc in Mechanical Engineering Thesis: "*Analytical and numerical study of the optimal placement of piezoelectric actuators on tapered beams.*"

Honour & Awards

2026	Robotics Travel Awards, Winner of the Travel award Issued by MDPI Robotics
2024	Rome Technopole Award (NextGenerationEU): "*Awarded for the "ECS000024 Rome Technopole Project" (CUP B83D21014170006).*"
2022-2023	Scholarship (Youth Fund for Year 2023) given by University of Rome Tor Vergata: "*Awardee for merit-based incentives aimed at first-year students who are enrolled in the 2022/2023 academic year*."
2021-2022	Scholarship (Academic Achievement) given by Roma Tre University: "*To students who have attained optimal outcomes related to the alignment of the study program and the grades achieved*."

Patents

2025	A. Perini, M. Russo, "Esoscheletro per mano e relativo metodo di fabbricazione," "*Hand exoskeleton and related fabrication methods*," Italian Patent IT 102025000009261, filed 24/04/2025.

Technical skills

Software & Hardware	MATLAB & Simulink, Autodesk Inventor, LATEX, MS Word, MS Excel, MS Powerpoint, Wolphram Mathematica, Fusion 360, ANSYS fluent, COMSOL Multiphysics, Affinity Designer, C, C++. 3D Printing, Sewing Machinist.
Languages	Italian: Native speaker English: Full Professional Proficiency (B2/C1)

Experience

PhD R&D Mechanical Design Intern | Sensor Medica, Italy (Sept 2026 – Present). Spearheading the mechanical design of a robotic gripper and driving the automation of a complete robotic cell to optimize production workflows.

Visiting PhD period (six months) | Technical University of Munich (TUM). Munich Institute of Robotics and Machine Intelligence (MIRMI) – ARIES lab.

Admission committee assistant | University of Rome Tor Vergata. Assessment & Grading: evaluated and scored entrance examinations for the Engineering Science program, ensuring compliance with academic standards.

Organization committee PhD EXPO | University of Rome Tor Vergata. Chairperson for the session: Pitch 3 Minute Thesis (3MT).

Student Representative for the IEEE Robotics & Automation Society (RAS) | Technical Committee on Mechanisms and Design for the 2026–2027 term.

Selected as entry: 2025 James Dyson Award | "TEXO, A lightweight, accessible glove exoskeleton, fully made of sustainable textiles"

Selected Bibliography

Perini, A., Piratoni, M., Tricomi, E., Russo, M., Masia, L. (2026). Barometric fusion for continuous variable assistance in a soft knee exoskeleton. IEEE RA-L (in prep.)

Perini, A., Russo, M., Hasan, M. I., Secciani, N., Ridolfi, A., Dragusanu, M. (2026). Towards reproducible metrics and benchmarks for soft hand exoskeletons: an open benchmark for grasping characterization. Advance Intelligent Systems, WILEY (under review)

Perini, A., & Russo, M. (2026). *From parametric patterns to tailored textile gloves for soft tendon-driven hand exoskeletons*. IEEE T-RO (under review)

Beyki, N., **Perini, A.**, Molinaro, L., Russo, M. (2026). *Muscle activation detection with surface electromyography: a combined signal processing and machine learning approach*. International Journal of Advanced Robotic Systems. DOI: https://doi.org/10.1177/17298806261446421.

Scuderoni, H., **Perini, A.**, Russo, M. (2026). *Design of a single-input, five-output differential actuation unit for underactuated hands*. 2026 IEEE International Conference on Robotics & Automation (ICRA), (Accepted).

Perini, A., & Russo, M. (2026). *Underactuation in Tendon-Driven Hand Exoskeletons*. In *International Conference on Motion Assisting Devices for Limbs* (pp. 230-244). Springer, Cham. DOI: https://doi.org/10.1007/978-3-032-15271-8_13.

Perini, A., & Russo, M. (2025, April). *Design of an adaptive tendon-driven hand exoskeleton for stroke rehabilitation*. In *2025 IEEE 8th International Conference on Soft Robotics (RoboSoft)* (pp. 1-8). IEEE. DOI: 10.1109/RoboSoft63089.2025.11020897